

PENGYU WANG

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EDUCATION

University of Washington, Michael G. Foster School of Business	Seattle, WA, USA
<i>Ph.D. in Operations Management</i>	Jan. 2023 – expected 2027
Advisors: Prof. Shi Chen, Prof. Yong-Pin Zhou, Prof. Masha Shunko, Prof. Mingwen Yang	
University of Washington, Department of Economics	Seattle, WA, USA
<i>Ph.D. in Economics</i> (program; transitioned to OM)	Sept. 2020 – Dec. 2022
University of Wisconsin–Madison, Department of Economics	Madison, WI, USA
<i>Master of Science in Economics</i>	Aug. 2018 – June 2020
Western University	London, Ontario, Canada
<i>Bachelor of Arts, Honors Specialization in Economics</i>	Sept. 2014 – June 2018

RESEARCH INTERESTS

Topics: Sustainable operations, Innovative value chain, Platform operations, AI infrastructure operations, Global supply chain challenges, Semi-conductor industry and supply chain.

Methods: Stochastic optimization, Game theory, Data analysis & Causal inferences, Mean-field games and control, Reinforcement learning.

RESEARCH

(†: Co-first authors).

Working Papers

1. **Wang, P.**, Chen, S., Hou, X. (2026). *Collective ESG Investment: Mitigation, Greenwashing, and Dynamic Strategic Interactions*. Reject & Resubmit at *Manufacturing & Service Operations Management* (Apr, 2026), Resubmitted (Aug, 2026).
 - Presented at INFORMS 2025.
 - *Scheduled presentation at INFORMS 2026. **Date:** Tuesday, November 3, 4:15 PM-5:30 PM. **Room:** Moscone South-70 (Lower Mezz).
 - We build a dynamic stochastic framework to study how heterogeneous firms make collective ESG investment decisions, including greenwashing and substantive mitigation, when their actions are strategically interdependent and interact with an ESG-aware financial market.
2. Li, Y.[†], **Wang, P.**[†], Zhou, Y., Shunko, M. *Can Less Be More? A Hybrid Solution to Print-on-Demand Platforms*. Reject & Resubmit at *Manufacturing & Service Operations Management* (Nov, 2025)

- Accepted by and presented at MSOM conference 2024.
 - Studies hybrid make-to-stock / make-to-order (MTS/MTO) production and assortment regulation on print-on-demand platforms. Develops analytical models and contracts to align incentives between platforms and content creators.
3. **Wang, P.[†]**, Zhang, J.[†], Chen, S., Yang, M. (2026). *Platform-Empowered Digital Finance and Traditional Investing: Spillover Effects through Financial Literacy and Risk Tolerance*. Under Review at *Manufacturing & Service Operations Management* (Aug, 2026).
 - *Accepted by and scheduled talk at CIST conference 2026, and DSI conference 2026.
 - This paper studies whether digital finance substitutes for traditional channels or broadens market participation. Using individual-level investment data, we conceptualize digital finance as a gateway that reduces participation frictions, facilitates experiential learning, and promotes migration into adjacent financial markets.
 4. **Wang, P.**, Zhou, Y., Shunko, M. *Marketplace Governance and Production Strategy: Decentralized Assortment and Hybrid Fulfillment in Creator Economy*. In preparation for submission.
 - * Scheduled talk at DSI 2026 conferece.
 - We study how creator portfolios are formed in digital marketplaces and the resulting production challenges for platforms. The separation between assortment creation and production decisions creates a need for marketplace governance, which platforms can address through market instruments that reshape creator portfolios and improve performance.

Work in Progress

5. **Wang, P.**, Chen, S. (2026) *A Two-Layer Hybrid Graph Mean-Field Game for Distributed LLM Inference Demand*.
 - Studies the offloading policy and pricing scheme in a data center network that provides AI-computing power.
6. **Wang, P.**, Chen, S. (2026) *Global Supply Chain Networks, Dynamics, and Disruption*.
 - Build a theoretical framework to uncover how global supply chain shocks penetrates through the supply network and how firms' endogenous R&D investments accelerate or contain such effects.
7. **Wang, P.**, Zhang, J., Chen, S. (2026). *Who Are the Winners? Business Opportunity Management in High-Tech Industry*.
 - Leverage a unique semi-conductor industry leading company's data to explore how such high-tech company explores business opportunity and foster future business partnership network.

Proceedings

8. **Wang, P.** (2017) *The Causal Relationship between Socioeconomic Status and Smoking Behavior: Evidence from Canada*. Proceedings in Advances in Economics, Business & Management.
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CONFERENCES & INVITED TALKS

A Two-Layer Hybrid Graph Mean-Field Game for Distributed LLM Inference Demand:
Decision Science Institute 2026, San Francisco, CA (scheduled)

Marketplace Governance and Production Strategy: Decentralized Assortment and Hybrid Fulfillment in Creator Economy:
Decision Science Institute 2026, San Francisco, CA (scheduled)

Collective ESG Investment: Mitigation, Greenwashing, and Dynamic Strategic Interactions:
INFORMS Annual Meeting 2026, San Francisco, CA (scheduled)
POMS Annual Meeting 2026, Reno, NV:
POMS-HK 2026, Shenzhen, China
INFORMS Annual Meeting 2025, Atlanta, GA

Can Less Be More? A Hybrid Solution to Print-on-Demand Platforms:
INFORMS Annual Meeting 2024, Seattle, WA
MOSM 2024 Annual Meeting 2024, Twin-City, MN

Others talks and experiences:
MSOM 2026 Annual Meeting, Boston, MA
SICS Annual Meeting 2025, Berkeley, CA
INFORMS Annual Meeting 2023, Phoenix, AZ

Society Services

Session chairs: INFORMS 2024, MSOM 2024, DSI 2026, INFORMS 2026

TEACHING EXPERIENCE

Invited Instructor: (Evaluations out of 5.0).

IGB 250 Winter 2026, Global Supply Chain: 4.5

IGB 250 Summer 2026, Global Supply Chain: 4.6

Teaching Assistance:

SCM 530A Autumn 2023, Operations Management And Process Analysis: 4.6

SCM 513 Autumn 2023, Managing And Mining Big Data: 4.5.

BUS AN 516 A/B Winter 2024, Operations Research Data Analytics: 4.2.

OPMGT 520A Spring 2024, Operations Management: 4.3

EMBA 533A Winter 2025, Operations Management And Systems Analysis: 4.5

QMETH 505A Spring 2025, Decision Modeling: 4.1;

SCM 520A Spring 2025, Managing Supply Chain Projects: 4.2.

REFERENCES

Prof. Shi Chen

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